

Advanced Trigonometry

ANSWER KEY



Trigonometric identities: Pythagorean and reciprocal

- 1 Given $\cos \theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin \theta$ and $\tan \theta$.
 $\sin^2 \theta = 1 - \frac{16}{25} = \frac{9}{25}$. In the fourth quadrant, sine is negative: $\sin \theta = -\frac{3}{5}$.
 $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-3/5}{4/5} = -\frac{3}{4}$.
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Sum and difference identities

- 2 Use the sum identity to find the exact value of $\sin(75^\circ)$. using $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$.
 $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.
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Double angle identities

- 3 Given $\sin \theta = \frac{3}{5}$ and θ in the first quadrant, find $\sin(2\theta)$ and $\cos(2\theta)$.
 $\cos \theta = \frac{4}{5}$ (first quadrant, positive). $\sin(2\theta) = 2\sin \theta \cos \theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.
 $\cos(2\theta) = 1 - 2\sin^2 \theta = 1 - 2\left(\frac{9}{25}\right) = 1 - \frac{18}{25} = \frac{7}{25}$.
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Graphs and transformations of trigonometric functions

- 4 State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.
Amplitude: 2. Period: $\frac{2\pi}{3}$. Phase shift: solve $3x - \frac{\pi}{2} = 0$, giving $x = \frac{\pi}{6}$ (shift right by $\frac{\pi}{6}$).
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Inverse trigonometric functions

- 5 Find the exact value of $\arcsin\left(\frac{1}{2}\right)$.
 $\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$ (or 30°), since $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$ and $\frac{\pi}{6}$ is in the range of arcsine.
- 6 Find the exact value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$.
 $\arccos\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}$ (or 135°), since $\cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$ and $\frac{3\pi}{4}$ is in the range of arccosine.
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Trigonometric equations with multiple solutions

- 7 Solve $2\cos^2 \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
 $\cos^2 \theta = \frac{1}{2}$, so $\cos \theta = \pm \frac{\sqrt{2}}{2}$. Solutions: $\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$.
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Applications: Law of Sines and Law of Cosines together

- 8 Two ships leave a port at the same time, one traveling at 20° and the other at 80° (measured from north), each at 15 mph. Find the distance between the ships after 2 hours, to the nearest mile, using the Law of Cosines.

Each ship travels $15(2) = 30$ miles. The angle between their paths: $80^\circ - 20^\circ = 60^\circ$. Using the Law of Cosines: $d^2 = 30^2 + 30^2 - 2(30)(30)\cos 60^\circ = 900 + 900 - 1800(0.5) = 1800 - 900 = 900$.
 $d = \sqrt{900} = 30$ miles.

Polar coordinates

- 9 Convert the polar point $(4, \frac{\pi}{3})$ to rectangular coordinates.
 $x = r\cos\theta = 4\cos(\frac{\pi}{3}) = 4(\frac{1}{2}) = 2$. $y = r\sin\theta = 4\sin(\frac{\pi}{3}) = 4(\frac{\sqrt{3}}{2}) = 2\sqrt{3}$. Rectangular coordinates: $(2, 2\sqrt{3})$.

A multi-part trigonometric identity verification problem

- 10 This question has two parts. (a) Use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan\theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan\theta = 3$.
(a) $\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$. Dividing numerator and denominator by $\cos^2\theta$:
 $\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}$. (b) $\tan(2\theta) = \frac{2(3)}{1 - 9} = \frac{6}{-8} = -\frac{3}{4}$.

Half-angle identities

- 11 Use the half-angle identity to find $\cos(15^\circ)$, given $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.
Half-angle identity: $\cos(\frac{\theta}{2}) = \sqrt{\frac{1 + \cos\theta}{2}}$. With $\theta = 30^\circ$:
 $\cos(15^\circ) = \sqrt{\frac{1 + \frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2 + \sqrt{3}}{4}} = \frac{\sqrt{2 + \sqrt{3}}}{2}$.

Solving trigonometric equations that factor

- 12 Solve $2\sin^2\theta - \sin\theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
This factors as a quadratic in $\sin\theta$: $(2\sin\theta + 1)(\sin\theta - 1) = 0$, so $\sin\theta = -\frac{1}{2}$ or $\sin\theta = 1$. For $\sin\theta = 1$: $\theta = 90^\circ$. For $\sin\theta = -\frac{1}{2}$: $\theta = 210^\circ$ or $\theta = 330^\circ$. All solutions: $90^\circ, 210^\circ, 330^\circ$.

Area of a triangle using trigonometry

- 13 Find the area of a triangle with sides 8 and 12, and included angle 50° , using $\text{Area} = \frac{1}{2}ab\sin C$.
 $\text{Area} = \frac{1}{2}(8)(12)\sin(50^\circ) \approx \frac{1}{2}(96)(0.766) \approx 36.8$.

Simplifying trigonometric expressions using identities

- 14 Simplify $\sin^2 \theta + \cos^2 \theta + \tan^2 \theta$ in terms of $\sec \theta$.

Since $\sin^2 \theta + \cos^2 \theta = 1$, the expression becomes $1 + \tan^2 \theta$. Using the Pythagorean identity $1 + \tan^2 \theta = \sec^2 \theta$, the simplified expression is $\sec^2 \theta$.

A multi-part navigation and triangulation problem

- 15 This question has three parts. A hiker walks 5 miles due east, then turns and walks 7 miles at a bearing of 40° north of east. (a) Find the angle between the two legs of the hike (measured as the interior angle at the turning point). (b) Use the Law of Cosines to find the straight-line distance from the start to the end point, to the nearest tenth of a mile. (c) Explain why this distance is less than the total distance walked (12 miles).

(a) The second leg deviates 40° from due east, so the interior angle at the turn (between reversing the first leg's direction and the second leg) is $180^\circ - 40^\circ = 140^\circ$. (b)

$$d^2 = 5^2 + 7^2 - 2(5)(7)\cos(140^\circ) = 25 + 49 - 70(-0.766) \approx 74 + 53.6 \approx 127.6.$$

$d \approx \sqrt{127.6} \approx 11.3$ miles. (c) The straight-line distance is always less than or equal to the sum of the individual leg lengths (a consequence of the triangle inequality) — walking in two different directions doesn't add distance as efficiently as walking directly toward the destination, since the path bends rather than going straight.

Verifying a trigonometric identity

- 16 Verify the identity $\frac{1 - \cos^2 \theta}{\sin \theta} = \sin \theta$.

Using the Pythagorean identity, $1 - \cos^2 \theta = \sin^2 \theta$. Substituting: $\frac{\sin^2 \theta}{\sin \theta} = \sin \theta$ (for $\sin \theta \neq 0$), confirming the identity holds.